

Matthew Pinto

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Summary:

Motivated Biocomputational Engineering student with strong interdisciplinary experience across wet-lab experimentation, computational modeling, and machine learning. Skilled in integrating experimental data with analytical and computational pipelines. Demonstrated ability to contribute to research leading to peer-reviewed publication. Seeking opportunities in research, biotechnology, computational biology, and interdisciplinary engineering.

Education:

University of Maryland College Park (Universities at Shady Grove Campus)

Coursework toward Bachelor of Science in Biocomputational Engineering (Expected September 2025 - May 2027)

Relevant Courses: Synthetic Biology, Python for Data Analysis, Machine Learning for Data Analysis, Applied Linear Systems and Differential Equations, Fluid Dynamics, Biomolecular Thermodynamics, Algorithms, C++, Quantitative Cellular and Molecular Biology, Molecular Techniques Laboratory

Syracuse University

Coursework toward Bachelor of Science in Biomedical Engineering (Aug 2023 - May 2025)

Relevant Courses: Ecology and Evolution, Engineering Statistics, Bioengineering Methods, Biological Principles for Engineers, Quantitative Physiology, Statics, Differential Equations and Linear Algebra, Organic Chemistry

Publications

- Zhang, K., Seitz, M. P., Pinto, M., Eghan, W. O.-A., & Jain, E. (2026). Ice Templated PEG–Alginate Double-Network Cryogels with Tunable Mechanics and Degradation for Soft Tissue Engineering. *Gels*, 12(6), 533. <https://doi.org/10.3390/gels12060533>
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Technical Skills

Wet Lab: Pipetting, Aseptic Technique, Cellular and Tissue Culture, PCR, Gel Electrophoresis, Spectrophotometry, Rheology, Compression, Freezer-Dryer, Degradation, Swelling

Dry-Lab: Python, Excel, Machine-Learning Analytics, Data Engineering, Quantitative Modeling, C++

Other: Engineering Statistics, Scientific Comprehension and Communication

Technical Experience:

Research Assistant — University of Maryland College Park

(July 2026-Present)

Programming Python scripts using NumPy to process 3D medical image volumes and manipulate voxel arrays.

- Implementing 3D dilation and spatial transformation algorithms to model vascular geometry changes in target Regions of Interest (ROI).
- Evaluating function robustness by testing boundary conditions, voxel-overwrite behavior, and spatial alignment across multi-layer tissue geometries.

GxP Systems Engineering Intern — Millipore Sigma

(June 2026 - August 2026)

Developed PowerShell automation that reduced manual audit-trail review length by over 90%, improving inspection readiness and reviewer throughput.

Built and evaluated machine learning models to identify drivers of protracted audit-trail length; performed structured data collection, preprocessing, and feature analysis.

Generated synthetic audit-trail datasets to test and validate PowerShell automation logic, ensuring consistent behavior across edge cases.

Executed data cleanup and standardization in Blue Mountain Ridge Asset Manager, enabling automatic scheduling and reducing delays in audit-trail review workflows.

Collaborated with QA, Systems Engineering, and Validation teams to align automation and ML outputs with GxP and data-integrity expectations.

Applied Lean Six Sigma DMAIC principles; earned White Belt and gained Yellow Belt-level experience through operational-excellence projects.

Practiced GxP-compliant system handling, including secure access management, audit-trail integrity, and adherence to safety and data-security protocols.

Research Assistant (November 2024 - May 2026) — Syracuse University

(November 2024 - July 2026)

Conducted research on biomaterial scaffolds for cartilage tissue engineering, focused on enhancing mechanical resilience and cellular compatibility

Synthesized and optimized biopolymer-based cryogels designed for injectable or load-bearing applications in regenerative medicine

Utilized Python and Excel and other various software programs for data analysis and tests including ANOVA, post-hoc tukey, linear regression, and correlation.

Utilized compression testing, rheometry, and degradation analysis to evaluate mechanical and viscoelastic behavior

Performed cell seeding, viability assays, and fluorescence microscopy to assess biological response and matrix infiltration

Operated instrumentation including mechanical testers, freeze dryers, incubators, and fluorescence microscopes to support data collection and scaffold characterization

Presented progress reports and contributed to data interpretation and experimental planning within a multidisciplinary research team

Findings demonstrated strong potential for translational applications in soft tissue repair and regenerative therapies

Invent@SU Participant — Syracuse University

(May 2024 - June 2024)

Founded prototype for wheelchair docking system eliminating back and forth shifting of current options on market

Tested, troubleshooted, and adapted 10+ versions of prototype to fit needs of client

Crafted and resourced a functioning electromagnetic clamping docking system throughout a 5 week timespan

Developed a complete business plan and a white paper to enter multiple business and invention competitions and placed second in one

Conducted extensive patent research to differentiate product from other similar ones on market

Engineering Projects

Flow Cytometry Analysis

Investigated the impact of salivary gland extract (SGE) and antibody-dependent enhancement (ADE) on dengue virus infection and immune cell recruitment in mouse dermal tissue using FlowJo for flow cytometry analysis.

Applied sequential gating strategies ascertain recruitment dynamics under various conditions: steady state, mosquito salivary gland extract (SGE), primary dengue infection, ADE, and combined ADE + SGE and identified increased neutrophil and myeloid cell proportions in dermis with highest levels in combined ADE + SGE + dengue virus group.

Interpreted qualitative trends in immune cell populations due to lack of biological replicates.

Contributed to understanding of dengue pathogenesis and immune modulation by mosquito saliva and enhancing antibodies.

Analyzed peer-reviewed articles to learn more about different immune cells' functions.

Real-Time Quantitative Polymerase Chain Reaction (RT-qPCR) Analysis

Engineered an analysis pipeline using RT-qPCR datasets to quantify gene expression changes in glioblastoma signaling networks across various treatment and microenvironmental co-culture conditions.

Characterized the transcriptional impact of the Hedgehog inhibitor cyclopamine, measuring downstream dysregulation in apoptosis, differentiation, sonic hedgehog pathway, and cell-cycle related genes.

Applied rigorous data normalization and statistical t-tests on biological triplicate samples to evaluate cross-experimental variance and validate published manuscript values for fold change and fold regulation.

Discovered that the tumor microenvironment (astrocytes) actively modulates tumor therapeutic response, leading to a significant alteration in SHH pathway expression and cell dormancy markers.

PCR and Gel Electrophoresis

Developed an endpoint single-tube multiplex PCR pipeline to genotype human subjects for the ACTN3 "sprint" gene variant utilizing the r577x polymorphism.

Managed the sterile extraction of genomic DNA from biological samples and validated template purity using spectrophotometric quality checks.

Adapted and optimized a 35-cycle thermal profiling protocol using a four-primer assay to simultaneously amplify control and allele-specific DNA fragments.

Engineered clear structural readouts by performing agarose gel electrophoresis to cross-examine and visualize experimental band sizes against a DNA ladder and determine genotypes for multiple participants

Effects of Caffeine on Cell Viability

Managed an in vitro cell culture study investigating the cytological stress response and viability impact of caffeine exposure on HEK293 mammalian cell lines.

Engineered an aseptic preparation method involving the centrifugation and filtration of beverage samples to create sterile liquid treatments.

Supervised a 24-hour incubation timeline testing regular and decaffeinated formulations against controlled Eagle's Minimum Essential Medium (EMEM) environments.

Analyzed cellular confluency trends, total viable counts, and cytotoxicity percentages to mathematically define sample variance and growth thresholds.

Executed data processing algorithms to establish low-concentration exposures maintain >99% overall cell viability with no statistically significant cytotoxicity.

Machine Learning to Predict Breast Cancer Survivability

Engineered a predictive machine learning framework evaluating breast cancer survivability risks from multi-parametric clinical and demographic patient datasets in Python.

Performed data cleaning techniques and trained, tested, and cross-validated a variety of classification based models from Scikit-Learn libraries to determine the impact of different features on breast cancer survivability.

Measured different model evaluation metrics to conclude different classification algorithms perform better at different tasks.

Parametric Orthopedic Screw Generator

Built a Python-driven parametric design tool that automatically generates patient-specific orthopedic screws.

Integrated SolidWorks API to automate geometry creation based on anatomical parameters.

Demonstrated ability to combine computational modeling with practical engineering design.

Leadership:

Vice President of Financials, ΛΦΕ International Fraternity, Syracuse University (November 2024-June 2025)

Budgeted 5-figure expenses and income of the organization.

Created fundraisers and planned events while minimizing costs to raise.

Communicated with 30+members to ensure timely payments.

Constructed parliamentary rules and consequences for ensuring high-quality efforts and actions.

Awards and Honors:

Deans List Fall 2023 - Spring 2026

Invest in SUccess Scholarship Spring 2024-Spring 2025

ChABSA Scholarship Fall 2025-Spring 2027

2024 Invent@SU Competition 2nd Place

2025 Lambda Phi Epsilon Academic Excellence Award